

Chebyshev's Inequality

Chebyshev's Inequality: For any random variable X and any $a > 0$ $\Pr(|X - E[X]| \geq a) \leq \frac{\text{Var}(X)}{a^2}$

Proof: Apply Markov's Inequality to $Y = (X - E[X])^2$
 $E[Y] = E[(X - E[X])^2] = \text{Var}(X)$ and $|X - E[X]| \geq a \Leftrightarrow Y \geq a^2$ so
 $\Pr(|X - E[X]| \geq a) = \Pr(Y \geq a^2) \leq \frac{E[Y]}{a^2} = \frac{\text{Var}(X)}{a^2}$

Example: If we flip a fair coin 6 times and $X = \# \text{ of heads}$, what bound does Chebyshev's Inequality give on $\Pr(X \leq 1 \text{ OR } X \geq 5)$? $= P(|X - 3| \geq 2) = P(|X - E(X)| \geq 2)$

$\text{Var}(X) = \frac{6}{4} = \frac{3}{2}$. By Chebyshev's Inequality, $\Pr(|X - E[X]| \geq 2) \leq \frac{\text{Var}(X)}{2^2} = \frac{\frac{3}{2}}{4} = \frac{3}{8}$

True probability: $2P(X \geq 5) = 2 \cdot \frac{7}{64} = \frac{7}{32} \approx 21.9\%$ $= 37.5\%$

Exercise: Can you give an example where Chebyshev's Inequality is tight, i.e. $\Pr(|X - E[X]| \geq a) = \frac{\text{Var}(X)}{a^2}$?